

UNIT-1 PROBABILITY AND RANDOM VARIABLES

Probability

Experiment: A physical action with an expectation

Deterministic experiment

An experiment whose outcome or result can be predicted with certainty is called a deterministic experiment.

For example, if the potential difference E between the two ends of a conductor and the resistance R are known, the current I flowing in the conductor is uniquely determined by Ohm's law, $I = E/R$

Random Experiment

Although all possible outcomes of an experiment may be known in advance, the outcome of a particular performance of the experiment cannot be predicted owing to a number of unknown causes. Such an experiment is called a random experiment.

Whenever a fair 6-faced cubic dice is rolled, it is known that any of the 6 possible outcomes will occur, but it cannot be predicted what exactly the outcome will be, when the dice is rolled at a point of time.

Trial :

Each Performance of a random experiment is called trial. Example: Tossing a coin 6 times is 6 trial

Sample space:

The set of all possible outcomes which are assumed to be equally likely are called sample space.

Example: For an exam result, sample space $S = \{O, A+, A, B, B+, UA, U, WH\}$.

Event:

Any subset of the sample space S is called an event.

The null set ϕ and S itself are two events. ϕ is called the impossible event and S is called sure events or certain event.

Note:

1. In practice any statement of conditions that defines this subset is called an event

Finite and Infinite sample space:

A sample space having finite number of sample points is called a finite sample space.

A sample space which is not finite is called on infinite sample space.

Note:2. If the sample space S has n elements, then the total number of events is 2^n .

Discrete Sample space:

A sample space S which is finite or countably infinite is called a discrete sample space.

Example : $S = \{HH, HT, TH, TT\}$ - is finite

$S = \{1, 2, 3, 4, \dots\}$ - is countable infinite .

Both are discrete sample space.

Continuous sample space:

A sample space which is uncountably infinite number is called a finite sample space.

Example: If the experiment consists of measuring the lifetime of a component (in hours) then the sample space $S = \{t/0 \leq t < \infty\}$.

Exhaustive events:

The total number of possible outcomes in any trail is known as exhaustive events.

Example: In tossing a coin the possible outcomes are getting a head or tail. Hence we have 2 exhaustive events in throwing a coin.

Mutually exclusive events:

Two events A and B are said to be mutually exclusive if $A \cap B = \phi$.

In practical situations, this means that A and B cannot occur together.

Example: In $S = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2\}$, $B = \{3, 4\}$ exclusive events, since $A \cap B = \phi$.

In tossing a coin the events head or tails are mutually exclusive since both tail and head cannot appear in the same time.

Equally likely events:

Two events are said to be equally likely if one of them cannot be expected to preference to other

Example: In tossing a coin head or tail are equally likely events

Independent Events:

Two events are said to be independent when the actual happening of one does not influence in any way the happening of the others.

$$(i.e) \quad P(A \cap B) = P(A) \cdot P(B)$$

Example: The results obtained by throwing a die are independent of the results obtained by drawing a king from a pack of card.

Pairwise Independent:

Three events A,B,C are said to be pairwise independent if

$$P(A \cap B) = P(A) \cdot P(B); \quad P(A \cap C) = P(A) \cdot P(C); \quad P(C \cap B) = P(C) \cdot P(B)$$

Mutually independent:

Three events are said to be mutually independent when the actual happening of one does not influence in any way the happening of the others.

$$(i.e) \quad P(A \cap B) = P(A) \cdot P(B); \quad P(A \cap C) = P(A) \cdot P(C); \quad P(C \cap B) = P(C) \cdot P(B)$$

$$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$$

Different Approaches to probability:

There are essentially three different ways of defining probability. They are:

- (1) Classical (or a priori) definition
- (2) Relative frequency limit (or a posterior) definition
- (3) Axiomatic definition

Mathematical or A priori or Classical Definition of Probability

Let S be the sample space and A be an event (a sub-set of S consisting to possible outcomes) associated with a random experiment. Let n(s) and n (A) be the number of elements of S and A. Then the probability of event A occurring, denoted as P (A), is defined by,

$$P(A) = \frac{n(A)}{n(S)} \quad \begin{array}{l} \text{[Number of cases favorable to A]} \\ \text{[Exhaustive number of cases in S]} \end{array}$$

Example: The probability of getting an even number in the die tossing experiment is 0.5, as $S = \{1,2,3,4,5,6\}$, $E = \{2,4,6\}$, $n(S) = 6$ and $n(E) = 3$.

Axiomatic Definition of Probability

Let S be the Sample space and A be an event associated with a random experiment. Then the probability of the event A, denoted P(A), is defined as a real number satisfying the following axioms.

- (i) $0 < P(A) < 1$ [Probability value lies between 0 and 1]
- (ii) $P(S) = 1$ [probability of sure event is 1]
- (iii) If A and B are mutually exclusive events, $P(A \cup B) = P(A) + P(B)$
- (iv) For any sequence of mutually exclusive events $A_1, A_2, \dots, A_n \dots$

$P(A_1 \cup A_2 \cup \dots \cup A_n \dots) = P(A_1) + P(A_2) + \dots + P(A_n) + \dots$ [Probability of at least one event is equal to the sum of the probability of other events]

1.2 Conditional probability

The conditional probability is a measure of the Probability of an event occurring, given that another event (by assumption, presumption, assertion or evidence) is already known to have occurred. The formula is

$$P(A/B) = \frac{P(A \cap B)}{P(B)}, P(B) \neq 0.$$

1.3 Total Probability and Baye's Theorem

Total Probability: The events $B_1, B_2, B_3, \dots, B_n$ is a set of exhaustive events of a sample space S, such that $\{B_1, B_2, B_3, \dots, B_n\}$ be the partitions of a sample space S, and the happening of the event A from the sample space S is $P(A) = P(B_1)P(A/B_1) + P(B_2)P(A/B_2) + \dots + P(B_n)P(A/B_n)$

Bayes' Theorem: In the above condition, if A happened then, $P(A/B_j) = \frac{P(B_j)P(A/B_j)}{P(A)}$

This is for you to understand probability 24MA301 syllabus starts from the next page

Random variable

Random variable:

A random variable is a function from S to R. Suppose you flip a coin two times. This simple statistical experiment can have four possible outcomes: HH, HT, TH, and TT. Now, let the variable X represent the number of Heads that result from this experiment. The variable X can take on the values 0, 1, or 2. In this example, X is a random variable

Types of Random Variable:-

1. Discrete Random Variable
2. Continuous Random Variable.

Discrete Random Variable:- A Random variable 'X' which can take a finite or countably infinite number of values is called a discrete random variable.

Example:-1. The number showed when a dice is thrown, then $X = \{1, 2, 3, 4, 5, 6\}$

Example:-2. The number of alpha particles emitted by a radioactive source.

Probability mass function: $P[X = X_i]$

Let X be a discrete random variable with each possible outcome X_i , a number $p(x_i) = P[X = X_i]$ give the probability that the random variable X equals the value of X_i . The numbers $p(x_i), i=1, 2, \dots$ must satisfy the following conditions: (i) $p(x_i) \geq 0$ for all i (ii) $\sum p(x_i) = 1$

Probability Distributions:- The collection of pairs $\{x_i, p(x_i)\}; i = 1, 2, 3, \dots$ is called the probability distribution of the random variable 'X'.

$X = x_i$	x_1	x_2	x_3	
$P(X = x_i)$	$p(x_1)$	$p(x_2)$	$p(x_3)$	

Continuous Random Variable:- A random variable which can take all values (ie, infinite number of values) in an interval, is called a continuous random variable.

Example:- The length of item during which a vacuum tube installed in a circuit functions.

Probability density function: $f(x)$

Let X be a continuous random variable which take all the values in an interval. The corresponding probability function $f(x)$ is called probability density function (pdf) defined by (i) $f(x_i) \geq 0$ (ii) $\int_{-\infty}^{\infty} f(x) dx = 1$

Cumulative distribution function (C.D.F):

For discrete Random Variable $F(x) = P(X \leq x) = \sum P(x)$

For continuous Random Variable $F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$

Properties of distribution function -F(x)

1. $F(x)$ is non-decreasing 2. $0 \leq F(x) \leq 1$ 3. $F(-\infty) = 0 \Rightarrow P(X \leq -\infty) = 0$ 4. $F(\infty) = 1 \Rightarrow P(X \leq \infty) = 1$	5. $P(a \leq X \leq b) = F(b) - F(a)$ 6. $F(X_1) \leq F(X_2)$ if $x_1 \leq x_2$ 7. $\frac{d}{dx} [F(x)] = f(x)$
--	---

Mean (μ) (OR) Expectation	$E(X) = \sum xp(x)$	$E(X) = \int_{-\infty}^{\infty} xf(x) dx$
Variance	$Var(X) = E(X^2) - [E(X)]^2$	$Var(X) = E(X^2) - [E(X)]^2$

Note:-

1. $E(\text{constant}) = \text{constant}$

2. $E(ax + b) = aE(x) + b$
3. $E(x + a)^2 = E(x^2 + a^2 + 2ax) = E(x^2) + a^2 + 2a E(x)$
4. $\text{Var}(ax + b) = a^2 \text{Var}(x)$
5. $P(X = x_i) = P(x_i) \rightarrow$ Probability function (or) Probability density (or) Probability Mass Function (PMF)
6. Conditional Probability $P(A/B) = \frac{P(A \cap B)}{P(B)}$ (OR) $P(X \leq x_1 / Y \leq y_1) = \frac{P(X \leq x_1 \cap Y \leq y_1)}{P(Y \leq y_1)}$
7. If X and Y are two independent random variables, then $E(XY) = E(X)E(Y)$

Examples: Discrete random variable:

1. A discrete random variable X has the following probability distribution.

x	0	1	2	3	4	5	6	7
$P(X)$	0	K	2k	2k	3k	K^2	$2K^2$	$7K^2 + k$

Find (i) 'k' (ii) $P[x < 6]$ (iii) the smallest value of 'c' for which $p(x \leq c) > 1/2$. (iv) $P[1.5 < x < 4.5 / x > 2]$.

2. If $P(X=x) = \begin{cases} \frac{x}{10}; & x = 1, 2, 3, 4 \\ 0; & \text{otherwise} \end{cases}$ is the pmf, then find $P(X < 3), E[x/2]$?

3. A probability mass function is defined as $P(X=0) = 3C^2$, $P(X=1) = 4C - 10C^2$ and $P(X=2) = 5C - 1$. Find (i) the value of C, (ii) $P[0 < x < 2 / x > 0]$ (iii) The distribution function of X (iv) The largest value of X for which $F(x) < 1/2$ and (v) $\text{Var}(X)$

4. If the Random variable 'X' takes the values 1, 2, 3, 4 such that $2P(X=1) = 3P(X=2) = P(X=3) = P(X=4)$, find the Probability distribution and cumulative distribution function of X.

5. If the Random variable 'X' takes the values $-3, -2, -1, 0, 1, 2, 3$ such that $P(X=0) = P(X > 0) = P(X < 0)$ and $P(X=-3) = P(X=-2) = P(X=-1) = P(X=1) = P(X=2) = P(X=3)$ then find (i) Probability distribution (ii) distribution function (iii) mean of Y where $Y = X^2 + 2X$.

6. The probability function of an infinite discrete distribution is given by $P[X=j] = \frac{1}{2^j}, j = 1, 2, 3, \dots$. Find (i) Mean and Variance of the distribution. (ii) $P(X \text{ is even})$, $P(X \text{ is odd})$, $P(X \geq 5)$, $P(X \text{ is divisible by } 3)$.

7. A continuous R.V 'X' that can assume any value between $X=2$ and $X=5$ has the density function given by $f(x) = k(1+x)$. Find (i) $P(X < 4)$, $P(2 < X < 3)$, $P(X < 3 / X < 4)$, $P(X > 3)$ (iii) r^{th} moment and hence find the mean and variance.

8. If $f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ k(2-x), & 1 < x < 2 \end{cases}$ is the pdf, find (i) k (ii) $P[0.2 < x < 1.2]$ (iii) $P[0.5 < x < 1.5 / x > 1]$ (iv) $F(x)$

9. If the density function of a continuous R.V 'X' is given by $f(x) = \begin{cases} 2x, & 0 \leq x \leq b \\ 0, & \text{elsewhere} \end{cases}$ then for what value of b is f(x) a valid pdf? Also find the CDF.

10. If the random variable X is known to have a distribution function, $F(x) = u(x)[1 - e^{-\frac{x}{5}}]$ then find mean of X.

11. If $f(x) = \frac{k}{1+x^2}$ is the pdf, then find k, $F(x)$ and $P[x > 0]$?

12. If $f(x) = 3x^2$, $0 < x < 1$, then find a, b such that $P[x \leq a] = P[x > a]$ and $P[x > b] = 0.05$?
13. The p.d.f of a continuous R.V 'X' is $f(x) = ke^{-|x|}$. Find 'k' and $F(x)$.
14. The milage in thousands of miles which car owners get with a certain kind of tyre is a random variable having p.d.f $f(x) = \frac{1}{20}e^{-\frac{x}{20}}$ for $x > 0$ and $= 0$ for $x \leq 0$. Find the probabilities that one of these tyres will last (i) atmost 10000 miles (ii) anywhere from 16000 to 24000 miles (iii) atleast 30000 miles.
15. If $p(x) = \begin{cases} xe^{-x^2/2}; & x > 0 \\ 0; & x < 0 \end{cases}$, then prove that $P(x)$ is a pdf. Also find probability density function $P(x)$.

MOMENTS

Moments: The expected value of an integral power of a RV is called moment.

(i) **rth moment about origin (μ'_r):** $\mu'_r = E(X)^r = \begin{cases} \sum (x)^r p(x), & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} (x)^r f(x) dx, & \text{if } X \text{ is continuous} \end{cases}$

(ii) **rth moment about mean (μ_r):** $\mu_r = E(X - \bar{X})^r = \begin{cases} \sum (x - \bar{x})^r p(x), & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} (x - \bar{x})^r f(x) dx, & \text{if } X \text{ is continuous} \end{cases}$
for $r = 1, 2, 3, \dots$

Relationship between μ'_r & μ_r

- First moment (or) Mean about the origin is $E[X] = \mu'_1$
- Second moment about mean (or) Variance $\mu_2 = \mu'_2 - (\mu'_1)^2$
- Third moment about mean $\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3$

Examples:

1. Find the mean, variance, $E[2X + 1]$, $\text{Var}(2X + 1)$ of a discrete R.V 'X' given its distribution as follows.

X	1	2	3	4	5	6
F(x)	1/6	2/6	3/6	4/6	5/6	1

2. A continuous R.V 'X' has a pdf $f(x) = 2(1 - x)$, $1 > x > 0$. Find r^{th} moment and $E[(2x+1)^2]$

3. A continuous R.V 'X' has a pdf $f(x) = kx^2e^{-ax}$, $x > 0$, $a > 0$. Find 'k', first four moments about the origin and variance.

4. A continuous R.V 'X' has a pdf $f(x) = \begin{cases} x, & 0 < x < 1 \\ 2 - x, & 1 < x < 2 \end{cases}$. Find the r^{th} moment and hence find mean.

5. If the pdf of a RV X is $f(x) = \begin{cases} ce^{-ax}, & x \geq 0 \\ 0, & x < 0 \end{cases}$ Find the r^{th} moment and hence find mean.

6. The probability distribution function of a random variable X is given by $f(x) = \frac{4x(9-x^2)}{81}$, $0 \leq x \leq 3$. Find the mean, variance and third moment about the origin.

7. Find the mean and variance of a R.V 'X' having the distribution function $F(x) = 1 - \frac{xe^{-kx}}{k} - \frac{e^{-kx}}{k^2}$, $x > 0$.

MOMENT GENERATING FUNCTION (MGF)

Moment Generating Function (MGF) of a R.V 'X' is defined as $E[e^{tx}]$, where 't' is a real variable and denoted as $M_X(t)$ (or) $M(t)$. (ie) $M_X(t) = \begin{cases} \sum e^{tx} p(x), & X \text{ is discrete} \\ \int_{-\infty}^{\infty} e^{tx} f(x) dx, & X \text{ is continuous} \end{cases}$

Method of Finding Moments (μ_r')

MGF of a random variable 'X' about the origin:

$$E(e^{tx}) = E \left[1 + \frac{tx}{1!} + \frac{(tx)^2}{2!} + \frac{(tx)^3}{3!} + \dots + \frac{(tx)^r}{r!} + \dots \right] = 1 + \frac{t}{1!} E(X) + \frac{t^2}{2!} E(X^2) + \dots + \frac{t^r}{r!} E(X^r) + \dots$$

$$\text{Moments} = \mu_r' = \text{Coefficient of } \frac{t^r}{r!}$$

Another method to find moments:- $\mu_r' = \left\{ \frac{d^r [M_X(t)]}{dt^r} \right\}_{t=0}$

$$\text{First moment } (\mu_1'):- \text{ If } r = 1 \Rightarrow \mu_1' = \left\{ \frac{d[M_X(t)]}{dt} \right\}_{t=0} \text{ (OR) } M_X'(0)$$

$$\text{Second moment } (\mu_2'):- \text{ If } r = 2 \Rightarrow \mu_2' = \left\{ \frac{d^2 [M_X(t)]}{dt^2} \right\}_{t=0} \text{ (OR) } M_X''(0)$$

Properties of MGF :-

1. If the MGF of X is $M_X(t)$ and if $Y = aX + b$ then $M_Y(t) = e^{bt} M_X(at)$.
2. If X and Y are two independent random variables then $M_{X+Y}(t) = M_X(t) M_Y(t)$.

Examples:

1. If $M_X(t) = \frac{e^t}{3} + \frac{4e^{3t}}{15} + \frac{2e^{4t}}{15} + \frac{4e^{5t}}{15}$, find the probability distribution.
2. Find the probability distribution of the total number of heads obtained in 4 tosses of a balanced coin. Hence find the MGF.
3. Find the MGF of a R.V 'X' having the distribution function $F(x) = \begin{cases} x^3/6, & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$.
4. Find the MGF of the R.V 'X' whose probability function $P[X = x] = \frac{1}{2^x}, x = 1, 2, 3, \dots$. Hence find its mean and variance.
5. If $f(x) = \frac{1}{3} e^{-\frac{x}{3}}$ for $x > 0$ is the pdf of X. Find MGF and first four moments about the mean.
6. Find the MGF of $f(x) = \begin{cases} \frac{x}{4} e^{\frac{x}{2}}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$ and hence find μ_1', μ_2', μ_3' and μ_4' .

PROBABILITY DISTRIBUTIONS

- Discrete distributions
- Continuous distributions

Discrete distributions:-Binomial, Poisson & Geometric distribution

(1)Binomial Distribution:-

A random R.V 'X' is said to follow binomial distribution if it assumes only non – negative values and its probability mass function is given by $P(X = x) = nC_x p^x q^{n-x}$, Here $x = 0, 1, 2, 3, \dots, n$; p – probability of success; q – probability of failure ; $p + q = 1$.

Moment generating function (MGF): $M_x(t) = (q + pe^t)^n$

E(X)=Mean= np and **Var(x)=** npq

Examples:

1. The mean & variance of a Binomial Distribution is 8 & 6. Find the probability of getting at least 4 successes.
2. A coin is biased so that a head is twice as likely to appear as a tail. If the coin is tossed 6 times, find the probabilities of getting (i) exactly 2 heads (ii) at most 4 heads (iii) at least 3 heads (iv) between 1 & 3 heads (inclusive).
3. Out of 800 families with 4 children each, how many families would be expected to have (a) 2 boys and 2 girls (b) at least 1 boy (c) at most 2 girls and (d) children of both sexes. Assume equal probabilities for boys and girls.
4. Sport stories and financial reports, written by algorithms based on artificial intelligence, have become common place. One company fed its algorithm with box scores and play-by-play information and created over one million on-line reports of little league games in 2011. They now write many stories on the Big Ten Network site. Suppose that the algorithm, or robot reporter, typically writes proportion 0.65 of the stories on the site. If 15 new stories are scheduled to appear on a web site next weekend, find the probability that (a) 11 will be written by the algorithm. (b) at least 10 will be written by the algorithm (c) between 8 and 11 inclusive will be written by the algorithm.
5. If the probability of success is 1/100, how many trials are necessary in order that probability of at least one success is greater than $\frac{1}{2}$?
6. An irregular 6 faced die is such that the probability that it gives 3 even numbers in 5 throws is twice the probability that it gives 2 even numbers in 5 throws. How many sets of exactly 5 trials can be expected to give no even number out of 2500 sets?

7. Fit the binomial distribution for the following data

X	0	1	2	3	4	5	6	Total
F	5	18	28	12	7	6	4	80

8. A machine manufacturing screws is known to produce 5% defective. In a random sample of 15 screws, what is the probability that there are

- (i) Exactly 3 defectives.

(ii) Not more than 3 defectives.

(2)Poisson Distribution:-

Poisson A R.V. 'X' taking non – negative values is said to follow poisson distribution if its pmf is given by $P(x) = \frac{e^{-\lambda} \lambda^x}{x!}, x = 0, 1, 2, \dots$

- **Moment generating function (MGF):** $M_X(t) = e^{\lambda(e^t - 1)}$

$E(X) = \text{Mean} = \lambda$ and $\text{Var}(x) = \lambda$

Examples:

1. If X, Y are independent Poisson variates such that $P[X = 1] = P[X = 2]$, $P[Y = 2] = P[Y = 3]$ then find $V[X - 2Y]$.
2. In a book of 520 pages, 390 typographical errors occur. Assuming Poisson's law for the number of errors per page. Find the probability that a random sample of 5 pages will contain no error.
3. It is known that the probability of an item produced by a certain machine will be defective is 0.05. If the produced items are sent to the market in packets of 20, using Poisson distribution, find the number of packets containing at least, exactly and at most two defective item in a consignment of 1000 packets.
4. A car hire firm has two cars, which it hires out day by day. The number of demands for a car on each day is distributed as a Poisson distribution with mean 1.5. Calculate the proportion of days on which (i) some demand is refused (ii) neither car is used.
5. Messages arrive at a switch board in a poisson manner at an average rate 6 per hour. Find the probability for each of the following events: (1) exactly 2 messages within one hr, (2) no messages within one hr (3) at least 3 messages within one hr. (4) fewer than 4 messages
6. Ten percent of the tools produced in a certain company are defective. Find the probability that in a sample of 10 tools chosen at random, exactly 2 will be defective by using (i) binomial (ii) The poisson approximation to the binomial distribution.
7. The number of monthly breakdown of a computer is a random variable with an average breakdown of 1.8. Find the probability that a computer will function for a month: (i). Without breakdown, (ii). With at least one breakdown (iii) With only one breakdown.
8. The atoms of a radioactive element are randomly disintegrating. If every gram of this element, on average, emits 3.9 alpha particles per second, what is the probability that during the next second the number of alpha particles emitted from 1 gm is (i) At most 6. (ii) At least 6. (iii) At least 3 and at most 6.
9. Six coins are tossed 6400 times. Using the Poisson distribution, what is the approximate probability of getting 6 heads x times?
10. The number of typing mistakes that a typist makes on a given page has a Poisson distribution with a mean of 3 mistakes. What is the probability that she makes
 - (i) Exactly 7 mistakes.
 - (ii) Fewer than 4 mistakes.
 - (iii) No mistakes on a given page.

(3)Geometric Distribution:-

A R.V 'X' is said to have a geometric distribution if its pmf is $P[x = x] = q^{x-1}p$; $x=1, 2, 3, \dots$

Note:- . Another form of this distribution is $P[x = x] = q^x p$; $x = 0, 1, 2, \dots$

- **Moment generating function (MGF):** $M_X(t) = E(e^{tx}) = \frac{pe^t}{1-qe^t}$

$$E(X) = \text{Mean} = 1/p, \text{Var}(X) = \frac{q}{p^2}$$

- **Memory less property of Geometric distributions:**

If X has a geometric distributions, then for any two positive integers m and n, $P[X > m+n | X > m] = P[X > n]$

Examples:

1. If the probability that a target is destroyed on any one shot is 0.5, what is the probability that it would be destroyed on sixth attempt?
2. A die is cast until 6 appears. What is the probability that it must cast more than five times?
3. If the probability that an applicant for a driver's licence will pass road test on any given trial is 0.8, what is the probability that he will finally pass the test (i) on the fourth trial and (ii) in fewer than 4 trials.
4. A coin is tossed until the first head occurs. Assuming that the tosses are independent and the probability of a head occurring is p. Find the value of p so that the probability that an odd number of tosses are required is equal to 0.6. Can you find a value of p so that the probability is 0.5 that an odd number of tosses are required?
5. A and B shoot independently until each has its own target. The probability of their hitting the target at each shot is $3/5$ and $5/7$ respectively. Find the probability that B will require more shots than A.

Continuous distributions :- Uniform, Exponential and Normal distribution

(1) Uniform Distribution:-

A Continuous R.V 'X' is said to follow Uniform (or) Rectangular distribution over an interval (a,b) if its p.d.f is given by $f(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{otherwise} \end{cases}$. Here 'a' and 'b' ($b > a$) are the parameters of the distribution.

- **Moment generating function (MGF)** $M_X(t) = \frac{e^{bt} - e^{at}}{t(b-a)}$

$$E(X) = \left[\frac{(b+a)}{2} \right] \text{ and } \text{Var}(X) = \left[\frac{4b^2 + 4ab + 4a^2 - 3b^2 - 6ab - 3a^2}{12} \right]$$

Examples:

1. If 'X' is uniformly distributed with mean 1 and variance $4/3$, find $P(X < 0)$.
2. A Random Variable 'X' has an uniform distribution over the interval $(-3, 3)$. Compute (i) $P(X = 2)$ (ii) $P(X < 2)$ (iii) $P(|X| < 2)$ (iv) $P(|X - 2| < 2)$ (v) CDF of X (vi) Variance.
3. Buses arrive at a specified stop at 15 minutes interval starting at 7 a.m. (that is they arrive at 7, 7.15, 7.30, 7.45 and so on) If a passenger arrives at the stop at a random time that is uniformly distributed between 7 and 7.30 a.m., find the probability that he waits (i) less than 5 minutes for a bus and (ii) at least 12 minutes for a bus.

4. For the rectangular distribution $f(x) = \begin{cases} \frac{1}{2a}, & -a < x < a \\ 0, & \text{otherwise} \end{cases}$ show that the moments of odd order are zero and even order (μ_{2r}) are $\frac{a^{2r}}{2r+1}$.
5. Let X be uniformly distributed random variable in the interval (a,9)& $P[3 < x < 5] = 2/7$. Find the constant a, compute $P[|x-5| < 2]$

(2) Exponential Distribution:-

A continuous R.V 'X' defined in $(0, \infty)$ is said to follow an exponential distribution with parameter λ if its p.d.f is given by $f(x) = \lambda e^{-\lambda x}$, where $\lambda > 0$ and $0 < x < \infty$.

$$M_X(t) = \frac{\lambda}{\lambda - t}, E(X) = \text{Mean} \frac{1}{\lambda}, \text{Var}(x) = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}$$

Note:- Instead of λ if θ given $f(x) = \frac{1}{\theta} e^{-\frac{x}{\theta}}$.

• Memory less property of Geometric distributions:

If X has a Exponential distributions, then for any two positive integers m and n , then $P[X > m+n | X > m] = P[x > n]$

Examples:

- Suppose X has an exponential distribution with mean equal to 10. Find the value of k such that $P[X < k] = 0.95$
- The amount of time that a camera will run without having to be reset is a random variable having exponential distribution with $\theta = 50$ days. Find the probability that such a camera will (i) have to reset in less than 20 days (ii) not have to be reset in at least 60 days.
- A component has an exponential time to failure distribution with mean of 10,000 hours. (i) The component has already been in operation for its mean life. What is the probability that it will fail by 15,000 hours? (ii) At 15,000 hours the component is still in operation. What is the probability that it will operate for another 5000 hours.
- The time (in hours) required to repair a machine is exponentially distributed with parameters $\lambda = 1/2$. What is the probability that the repair time exceeds 2 hours? what is the conditional probability that a repair takes atleast 10 hours given that its duration exceeds 9 hours.
- The lifetime X of particular brand of batteries is exponentially distributed with a mean of 4 weeks. Determine
 - The mean and variance of X,
 - What is the probability that the battery life exceeds 2 weeks?
 - Given that the battery has lasted 6 weeks, what is the probability that it will last at least another 5 weeks?
- The life (in years) of a certain electrical switch has an exponentially distribution with an average life $1/\lambda = 2$. If 100 of these switches are installed in different systems, find the probability that the at most 30 fail during the first year?

(3) Normal Distribution:-

A continuous R.V 'X', assuming all real values in $-\infty < X < \infty$ has a Normal distribution if its p.d.f is of the form $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$. Here μ and σ^2 are the parameters of this distribution. Hence it is also written as $N(\mu, \sigma^2)$.

Standard Normal Variate and its Distribution:- If $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma}$ is called the standard normal variate. The p.d.f of Z is given by $\phi(z) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{z^2}{2}}$, $-\infty < Z < \infty$.

Examples:

- Find the probabilities that a random variable having the standard normal distribution will take on a value (i) between 0.87 and 1.28 (ii) -0.34 and 0.62 (iii) greater than 0.85 (iv) greater than -0.65
- If X is a normal variate with mean $\mu = 30$ and variance 100, then find (i) $P(X \leq 100)$ (ii) $P(X \leq 80)$ (iii) $P(65 \leq X \leq 100)$ (iv) $P(70 < X)$ (v) $P(85 \leq X \leq 95)$ (vi) $P(|X-90| > 15)$.
- Assume that the reduction of a person's oxygen consumption during a period of Transcendental Meditation (T.M) is a continuous random variable X which is normally distributed with mean 37.6 cc/min and S.D 4.6 cc/min. Determine the probability that during a period of T.M. a person's oxygen consumption will be reduced by (1) at least 44.5 cc/min (2) at most 35.0 cc/min (3) anywhere from 30.0 to 40.0 cc/min.
- The marks obtained by a number of students in a certain subject are assumed to be normally distributed with mean 65 and standard deviation 5. If 3 students are selected at random from this group, what is the probability that two of them will have marks over 70?
- In a normal distribution, 31% of the items are under 45 and 8% are over 64. Find the mean and variance of the distribution.
- The average percentage of marks of candidates in an examination is 42 with a standard deviation of 10. If the minimum marks for pass is 50% and 1000 candidates appear for the examination, how many candidates can be expected to get the pass mark if the marks follow normal distribution? If it is required, that double the number of the candidates should pass, what should be the minimum mark for pass?
- In a normal population with mean 15 and standard deviation 3.5, it is found that 647 observations exceed 16.25. What is the total number of observations in the population?